

# Impact response of pre-strained pure vanadium

Cite as: J. Appl. Phys. 131, 215905 (2022); doi: 10.1063/5.0092904

Submitted: 24 March 2022 · Accepted: 17 May 2022 ·

Published Online: 6 June 2022



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## ABSTRACT

The effect of modest, 0.6% and 5.5%, pre-straining on the impact response of 2 mm thick samples of annealed polycrystalline vanadium of commercial purity was studied in a series of planar impact tests. The loading of the samples by 0.5 mm thick copper impactors having velocities varying between 300 and 610 m/s was accompanied by continuous laser Doppler velocimetry of their rear surface. Based on the recorded velocity histories, the dynamic compressive  $\sigma_Y$  and tensile (spall)  $\sigma_{sp}$  strengths and the strength  $\sigma_Y^{sc}$  of vanadium in the shock-compressed state were determined. Adjacent to the impact surface part of the cross sections of the softly recovered samples, the number of twins  $N_{tw}$  per unit area was counted. It was found that the main parameter governing both the strength  $\sigma_Y$  of pristine (in the shock sense) material and that in the shock-compressed state,  $\sigma_Y^{sc}$ , was the initial dislocation density  $\eta_0$ . Moreover, the dislocation surplus caused by pre-straining was responsible for complete suppressing of twinning in the 0.6% and 5.5% pre-strained samples. In undeformed vanadium, the twinning was partially suppressed by the presence of impurity atoms which, however, did not affect the twinning stress, which was equal to approximately 0.7 GPa.

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## I. INTRODUCTION

As a metal with BCC lattice, vanadium is a relatively hard material characterized by high Peierls stress.<sup>1,2</sup> At the same time, its ductility is much higher and stacking fault energy and Peierls stress are lower than those of other BCC metals.<sup>1–4</sup> When loaded, either statically or dynamically at room temperature, vanadium undergoes plastic deformation carried, as in other BCC metals, by dislocations with Burgers vector  $\mathbf{b} = a/2\langle 111 \rangle$  ( $a = 0.303$  nm) gliding in  $\{110\}$  and  $\{211\}$  planes (e.g., Refs. 1 and 5–7). Moreover, recent studies of vanadium impact response show that twinning is an alternative to the glide mode of plastic deformation in vanadium at room temperature.<sup>8,9</sup> (Prior to these studies, the deformation twinning in vanadium was observed at temperatures not higher than 77 K.<sup>1,3,10–13</sup>) The twin planes in vanadium, as in other BCC metals, are  $\{211\}$  planes and, correspondingly, the twinning scheme in vanadium is of  $\{111\}\{211\}$  type. The existence of two paths, gliding and twinning, of plastic deformation implies either competition or interaction between them. In the case of routine quasi-static testing of such a two-path material at room temperature, the glide evidently prevails. Under a gradual increase of

loading, all sample (gauge) cross sections bear virtually the same stress. Since the shear stress corresponding to the onset of the glide,  $\tau_{CRSS}$ , is lower than that corresponding to the onset of twinning,  $\tau_{tw}$ , the triggering of the glide makes the twinning unachievable. At temperatures lower than 77 K, the situation changes due to different temperature dependencies of  $\tau_{CRSS}$  and  $\tau_{tw}$ ; the two values become closer and the twinning, although being usually preceded by some amount of glide, becomes possible. Such simultaneity of glide and twinning, however, leads to a substantial uncertainty of  $\tau_{tw}$  determination. Due to the high  $\tau_{tw}$ , the twinning cannot be started without a concentration of the shear stress, e.g., at the head of the dislocation pileup,<sup>14</sup> but the number of dislocations in the pileup cannot be controlled. As a result, the experimentally determined values of  $\tau_{tw}$  are characterized by substantial scatter.<sup>14–16</sup>

The situation changes when twinning is activated by planar impact loading, e.g., Refs. 8 and 9. The shear stress generated by the impact at the sample front surface may be high enough for simultaneous and independent activation in a pristine material of both glide and twinning, thus providing a “fair” competition between the mechanisms. In Refs. 8 and 9, intense twinning was

22 December 2025 18:07:02

observed adjacent to the impacted surface of the samples of pure annealed vanadium. This provided a motivation to study the effect of the change in the initial sample state on this glide-twinning competition. Generating a controlled amount of deformation twins in the sample assigned for the impact testing is hardly possible. On the contrary, a modest controllable sample pre-straining may be easily realized. Pre-straining of any metal at room temperature is usually associated with plastic deformation by the glide, i.e., by introducing into the material some surplus, with respect to undeformed material, of dislocations. In the present work, the effect of such pre-straining was studied in a series of planar impact tests in which undeformed and pre-strained (up to a residual strain of about 0.6% and 5.5%) 2 mm thick vanadium samples were softly recovered after impact loading by 0.5 mm thick copper impactors having velocities of about 300, 470, and 610 m/s, and then a metallographic examination was conducted.

Another motive for the present work was to examine the effect of such variations of initial samples' state on their dynamic tensile (spall) strength and their yield strength in the shock-compressed state. While the former is a routine result of planar impact tests,<sup>17</sup> estimating of the latter is more complicated. Based on findings reported in Ref. 18, the difference between two values of shear stress  $\tau(u_{p1})$  and  $\tau(u_{p2})$  corresponding to the particle velocities  $u_{p1}$  and  $u_{p2}$  in a simple rarefaction wave is equal to

$$\tau(u_{p1}) - \tau(u_{p2}) = \frac{3}{4}\rho_0 \int_{u_{p2}}^{u_{p1}} [a^2(u_p) - a_b^2(u_p)] \frac{du_p}{a(u_p)}, \quad (1)$$

where  $a(u_p)$  and  $a_b(u_p)$  are, respectively, Lagrangian phase velocity and Lagrangian bulk speed of sound,  $\rho_0$  is the initial material's density, and stress and strain are assumed to be positive in compression. If the particle velocity  $u_{p1}$  corresponds to the beginning of the quasi-elastic unloading and  $u_{p2}$  corresponds to the beginning of the plastic unloading (i.e., in the course of the unloading, the shear stress becomes zero at the Hugoniot hydrostat and then achieves the flow stress with an opposite sign), Eq. (1) actually determines the yield strength  $\sigma_Y^c$  of the shock-compressed material. To be applied to velocity histories recorded at the rear free surface of the shocked samples, Eq. (1) was simplified in Ref. 19 assuming, in particular, that Hugoniot  $P = \rho_0(c_0 + su_p)u_p$  and the unloading isentrope coincide in  $P - u_p$  coordinates<sup>20</sup> and that the Lagrangian bulk speed of sound may be expressed as  $a_b = c_0 + 2su_p$  (Ref. 21, p. 60). The approach suggested in Ref. 19 was successfully applied to determine the  $\sigma_Y^c$  of shock-compressed FCC aluminum,<sup>22</sup>

copper,<sup>23</sup> and HCP titanium<sup>24</sup> at room and elevated temperatures. It was found that at elevated temperatures in all studied metals, the values of  $\sigma_Y^c$  were substantially lower than the yield strength,

$$\sigma_Y = \frac{3}{2}\sigma_{HEL} \left(1 - \frac{c_b^2}{c_l^2}\right) \quad (2)$$

determined based on the material's Hugoniot elastic limit (HEL),

$$\sigma_{HEL} = \frac{1}{2}\rho_0 c_l u_{HEL}, \quad (2a)$$

where  $c_l$  and  $c_b$  are, respectively, longitudinal and bulk speeds of sound and  $u_{HEL}$  is the sample free surface velocity at HEL. (The room temperature values of  $\sigma_Y^c$  in Al<sup>22</sup> are virtually equal and in Cu<sup>23</sup> are slightly higher than  $\sigma_Y$ .) At the same time, in all the studied materials,  $\sigma_Y^c$  was found to grow with the increase of the peak loading stress. Since no such analysis was previously performed with BCC metals, it was proposed to apply it to the velocity histories recorded in the planar impact tests with undeformed and pre-strained vanadium samples.

## II. MATERIAL AND EXPERIMENTAL

### A. Preparation of the samples for impact tests

The billets for the preparation of vanadium samples were cut from a 6 mm thick plate of vanadium of 99.38% purity containing 0.15% Fe, 0.2% Al, 0.2% Si, and up to 0.01%, 0.001%, 0.03%, and 0.03% N, H, O, and C, respectively. The billets were annealed in a vacuum ( $10^{-4}$  Torr) furnace at 1000 °C for 1 h and cooled with the furnace. According to Refs. 8 and 9, such annealing is sufficient for complete removal of effects of rolling-related deformations. The samples,  $2 \times 20 \times 20$  mm<sup>3</sup> squares, were cut from the billets by electrical discharge machining (EDM) and ground to 0.2 mrad parallelism. Three groups of the samples were prepared. The first group, VA0 in follows, contained undeformed samples. The samples of two other groups, VA0.6 and VA5.5, were deformed, by applying uniaxial stress equal to 375 and 625 MPa, up to residual strains equal to 0.6% and 5.5%, respectively. One sample (not used in further experiments) was pre-strained up to a residual strain of 0.1% by applying uniaxial stress equal to 162 MPa. The latter value is considered as a quasi-static yield strength (proof stress)  $\sigma_Y^{0.1}$  of vanadium. Prior to the impact testing, the initial density  $\rho_0$  (Archimedes method) and longitudinal  $c_l$  and shear  $c_s$  speeds of sound (pulse-echo technique, 5 MHz transducers) were determined for each sample. The average, for each group, physical properties are listed in Table I, together with the estimated values of the bulk

TABLE I. Physical properties of three groups of vanadium samples.

| Sample group | $\rho_0$ ( $\frac{g}{cm^3}$ ) | $c_l$ ( $\frac{km}{s}$ ) | $c_s$ ( $\frac{km}{s}$ ) | $c_b$ ( $\frac{km}{s}$ ) | $E'$ (GPa) | $G$ (GPa) | $B$ (GPa) | $\nu$    |
|--------------|-------------------------------|--------------------------|--------------------------|--------------------------|------------|-----------|-----------|----------|
| VA0          | 6.11(1) <sup>a</sup>          | 6.09(2)                  | 2.81(2)                  | 5.17(3)                  | 228(1)     | 48.3(3)   | 163(2)    | 0.365(3) |
| VA0.6        | 6.11(1)                       | 6.08(2)                  | 2.82(2)                  | 5.14(3)                  | 226(1)     | 48.5(3)   | 162(2)    | 0.363(3) |
| VA5.5        | 6.11(1)                       | 6.10(2)                  | 2.81(2)                  | 5.17(3)                  | 226(1)     | 48.1(3)   | 163(2)    | 0.366(3) |
| Aver.        | 6.11 <sup>b</sup>             | 6.09                     | 2.81                     | 5.16                     | 226        | 48.3      | 163       | 0.365    |

<sup>a</sup>Digit in parenthesis corresponds to the uncertainty of the last digit of the property value.

<sup>b</sup>Average values of the bottom row are used hereafter as the properties of vanadium in different initial states.

22 December 2025 18:07:02

speed of sound  $c_b = \sqrt{c_l^2 - (4/3)c_t^2}$ , longitudinal  $E' = \rho_0 c_l^2$ , shear  $G = \rho_0 c_t^2$ , and bulk  $B = \rho_0 c_b^2$  moduli and Poisson's ratio  $\nu = (1 - 2G/E')/(2 - 2G/E')$ . As is apparent from Table I, the changes in the properties caused by pre-straining are modest.

One sample of each group was spared for metallographic (light and scanning electron microscopy) examination of the initial, prior to impact loading, material state. Cutting, polishing, and etching of these samples were the same as of the samples softly recovered after impact loading (see Subsection II C).

## B. Planar impact testing of the samples

Planar impact experiments with vanadium samples were carried out using a 50 mm smooth bore gas gun. The 2 mm thick plane-parallel samples were impact loaded by plane-parallel 0.5 mm thick copper impactors glued to a 5 mm thick polymethylmethacrylate substrate located at the front edge of a hollow aluminum projectile. The samples of each group were loaded by impactors having velocities  $300 \pm 10$ ,  $470 \pm 10$ , and  $610 \pm 10$  m/s measured by electrical charged pins. Consequently, the elastic compressive stress  $\sigma_{EL}$  generated by such loading at the copper/vanadium interface was equal to  $5.9 \pm 0.2$ ,  $9.3 \pm 0.2$ , and  $12.1 \pm 0.2$  GPa. [Elastic compressive stress  $\sigma_{EL} = Z_V Z_{Cu} v_0 / (Z_V + Z_{Cu})$  was determined based on impact velocity  $v_0$  and elastic impedances  $Z_i = \rho_0^i c_l^i$  of vanadium and copper.] In all impact tests, the impactor/sample misalignment did not exceed 0.5 mrad, providing uniaxial compression of the sample in the impact direction. During the tests, the velocity histories  $u(t)$  of the rear (unimpacted) sample surfaces were continuously monitored by VISAR (Velocity Interferometer System for Any Reflector<sup>25</sup>) with interferometer constant  $U_0 = 168$  m/s. After extracting the shock-loaded vanadium samples from the "rugs and oil"<sup>9</sup> catcher, the samples were cut along the centerline for postmortem metallographic examination. Taking into account the fact that the region of interest was the part of the sample cross section adjacent to the impacted surface and close (1–1.5 mm) to its center, no special efforts<sup>9,27–30</sup> for preventing the arrival of the release waves to the region were made. It should be noted that during implementation of Ref. 9, the waiver of the release-preventing measures was justified by an excellent coincidence of

the twinning count in trap-equipped 12.5 mm (trap outer diameter 23 mm) and in 20 mm trap-free vanadium samples recovered after shock loading by copper impactors having velocities of  $602 \pm 1$  and  $604 \pm 1$  m/s, respectively.

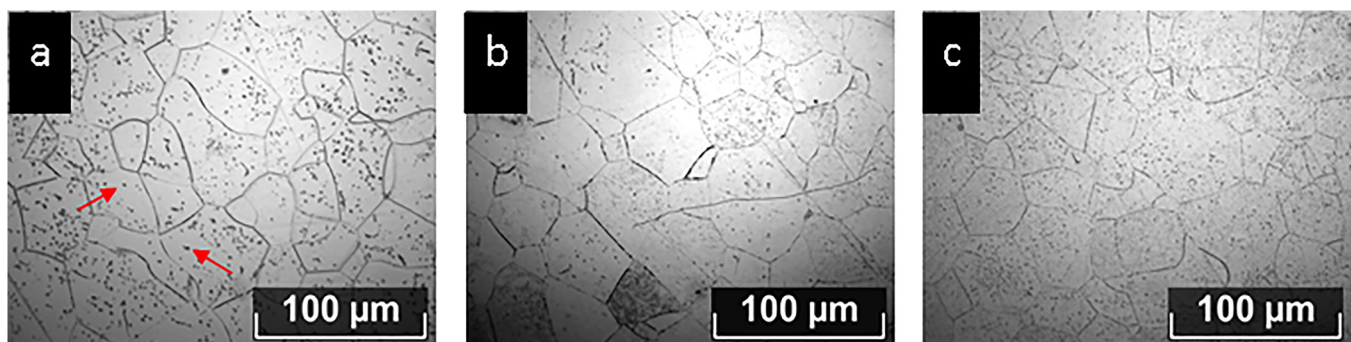
## C. Metallographic examination of the initial and postmortem sample states

Both unloaded and softly recovered shocked samples underwent the same stages of metallographic preparation as vanadium samples studied in Ref. 9. The samples were cut along the centerline, mounted in acrylic, ground on 1200, 2500, and 4000 grit SiC papers and polished, first, with a  $3 \mu\text{m}$  diamond paste and finally by  $0.04 \mu\text{m}$  colloidal silica. The polished samples were etched for 15–20 s with a solution containing 30 ml of  $\text{H}_2\text{O}$ , 30 ml of 65% nitric acid, 30 ml of 32% hydrochloric acid, and 15 ml of 40% hydrofluoric acid. Characterization was carried out on light microscope ZEISS Axiovert 25 equipped with a Paxcam3 camera and on a FEI Verios 460L high resolution SEM (Scanning Electron Microscope). The grain size was determined using Thixomet software<sup>31,32</sup> for image analysis. Due to the small ( $1\text{--}2 \mu\text{m}$ ) width of the deformation twins observed in the shocked samples, the twinned state of these samples was characterized by the number of twins per unit area,  $N_{tw}$ , as was done in Ref. 9. Unlike in Ref. 9, where  $N_{tw}$  was determined over a 2 mm wide strip along all cross-sectional centerlines, in the present study, the twin density was determined for the cross-sectional region bounded by a  $2 \times 0.2 \text{ mm}^2$  rectangle whose long side coincided with the impacted sample surface and whose centerline coincided with the centerline of the sample cross section. The correctness of  $N_{tw}$  determination was controlled by a selective recount of twins using SEM images of the samples' cross sections.

## III. EXPERIMENTAL RESULTS

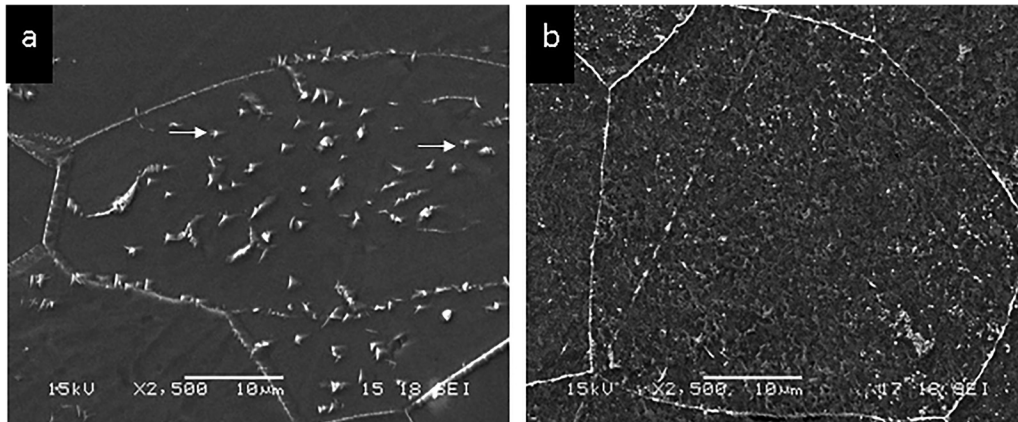
### A. Initial state of undeformed and pre-strained vanadium samples

Typical LM (Light Microscope) images of the cross sections of undeformed and pre-strained samples are shown in Fig. 1. The average grain size  $D$  is found to be virtually unaffected by



**FIG. 1.** Typical light microscope (LM,  $\times 500$ ) images of the cross sections of the VA0 (a), VA0.6 (b), and VA5.5 (c) samples. The arrows show two of multiple etch pits associated with the presence of dislocations.

22 December 2025 18:07:02



**FIG. 2.** SEM images of grains of similar size in the VA0 (a) and VA0.6 (b) samples. The arrows show two of multiple etch pits decorated by foreign (C and N) atoms.

pre-straining and equal to  $58 \pm 16$ ,  $61 \pm 17$ , and  $57 \pm 19 \mu$  for VA0, VA0.6, and VA5.5 samples, respectively. (In each case, approximately 150 grains were evaluated.) The number of dislocation etch pits in the grains of the undeformed samples seems to be greater than that in the grains of the pre-strained samples.

As is apparent from the SEM images in Fig. 2, the dislocation density  $\eta$  in the deformed samples is so high that neighboring individual etch pits become unresolvable. (Two similar grains with  $D \approx 60 \mu$  close to the average grain size were chosen for Fig. 2.) A determination of the dislocation density  $\eta$  based on the direct count of the etch pits in the grain of undeformed vanadium, Fig. 2(a), gives  $\eta \approx 1.5 \times 10^{11} \text{ m}^{-2}$ , the value close to the upper bound of the etch-pit method applicability.

In Fig. 2(b), only some of the pits (unmerged) may be resolved, but the presence of a dotted pits line (likely to be a dislocation pileup) allows one to conclude that dislocation paths  $L$  are of the order of  $D$ . In such a case, the value of the dislocation density may be estimated based on Orowan's equation  $\gamma = \eta b L$  (Ref. 26, p. 791), where  $\gamma$  is the plastic strain equal to a doubled pre-strain value, 0.006 or 0.055, and  $b = 0.262 \text{ nm}$  is the Burgers vector. Therefore, the dislocation densities in the pre-strained samples VA0.6 and VA5.5 are, respectively,  $\eta \approx 10^{12} \text{ m}^{-2}$  and  $\eta \approx 10^{13} \text{ m}^{-2}$ . The differences in the initial, prior to the impact loading, dislocation densities strongly affect both the impact response of the samples and their postmortem structure.

## B. Velocity histories of undeformed and pre-strained vanadium samples

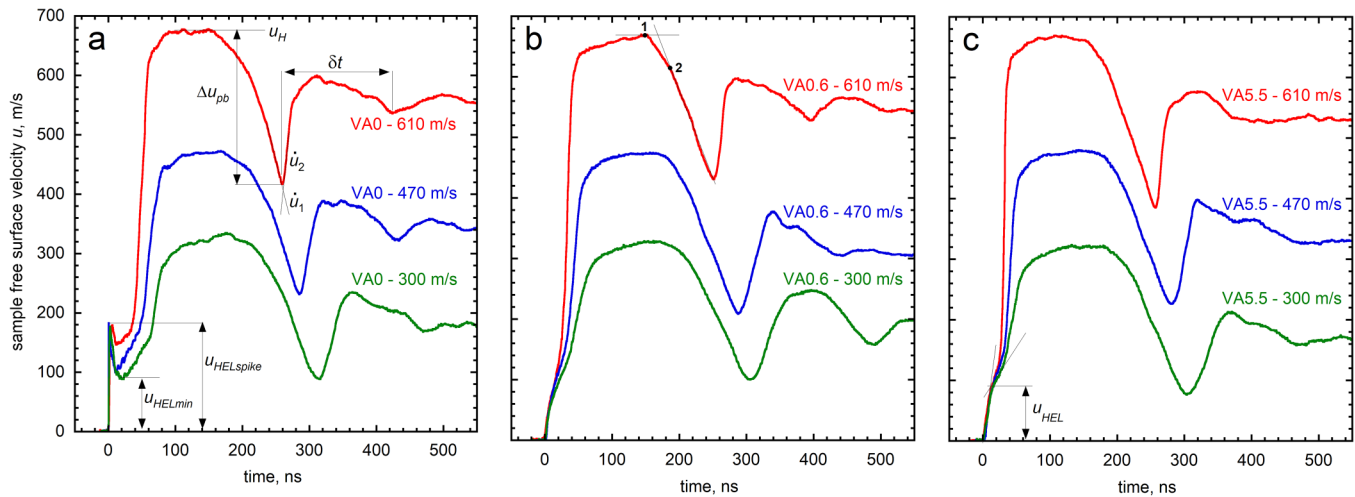
The velocity histories recorded with the differently pre-strained samples shock loaded by copper impactors having velocities of about 300, 470, and 610 m/s are shown in Fig. 3. All the waveforms contain elastic precursor waves with amplitude  $u_{HEL}$  followed by a plastic wave terminated at Hugoniot state  $u_H$ . Arrival of the unloading wave at the free sample surface results in its deceleration toward the velocity  $u = u_H - \Delta u_{pb}$ . The value  $\Delta u_{pb}$  is associated with the materials' dynamic tensile (spall) strength.

The shape of elastic precursor waves recorded with undeformed samples, Fig. 3(a), differs strongly from those obtained with pre-strained samples. The presence of a spike (followed by velocity minimum) at the precursor part of the waveform is usually attributed to fast generation and multiplication of the carriers of plastic deformation just behind the elastic wave front. A similar shape of elastic precursor wave recorded after loading 2 mm thick annealed vanadium samples by 0.5 mm copper impactors was observed in Ref. 8, Fig. 4(a). The samples studied in Ref. 8 contain less impurities (99.8% V against 99.38% in the present study) and, while annealed, were characterized by a much greater, of about  $180 \mu\text{m}$ , average grain size.

As apparent from Figs. 3(b) and 3(c), pre-straining changes substantially both the shape and the amplitude of the elastic precursor wave. The amplitude of the spikes obtained with the VA0 samples is about 180 m/s (100–150 m/s at minima). The precursor waves of the pre-strained samples are characterized by a rounded spike-less shape, while the precursors' amplitude drops threefold, to  $\sim 56 \text{ m/s}$ , after 0.6% pre-straining, and then, with 5.5% pre-straining, grows to 77–83 m/s. Moreover, while the rise time of the precursor front in the undeformed samples is of the order of magnitude of VISAR temporal resolution ( $\sim 1 \text{ ns}$ ), the rise time of the precursor front increases up to 7–8 ns after 0.6% pre-straining and up to 12–13 ns in 5.5% pre-strained samples. Figure 4(b) shows such changes in the case of 470-m/s impact loading. The rise times of the plastic waves demonstrate an opposite trend. The slopes of the plastic fronts increase with impact strength. The same is true for the maximum rate of compressive deformation  $\dot{\epsilon}_{comp}$  in a plastic wave determined based on its slope  $\dot{u}_{max}$ ,  $\dot{\epsilon}_{comp} = \dot{u}_{max}/2U_{pl}$ , where  $U_{pl} = c_l(1 - \Delta t/2t_0)/(1 + \Delta t/2t_0)$  is the velocity of the plastic wave in Lagrangian coordinates (Ref. 21, p. 47), and  $\Delta t$  is the time interval between the arrival of elastic and plastic waves at the sample free surface.

It is apparent from Fig. 4(c) that pre-straining leads to a substantial increase in the compressive strain rate. It is likely that the decrease in the dynamic viscosity of vanadium is caused by the dislocations' surplus produced by pre-straining. It should be





**FIG. 3.** Sample free surface velocity histories recorded after a shock loading of vanadium samples in undeformed (a) and pre-strained to residual strain 0.6% (b) and 5.5% (c) states. The velocities of copper impactors are included in the materials' designations. The arrows and points (1 and 2) show waveform parameters used in estimating  $\sigma_{HEl}$ , dynamic yield strengths  $\sigma_Y$  and  $\sigma_Y^{sc}$ , and dynamic tensile (spall) strengths of differently pre-strained vanadium.

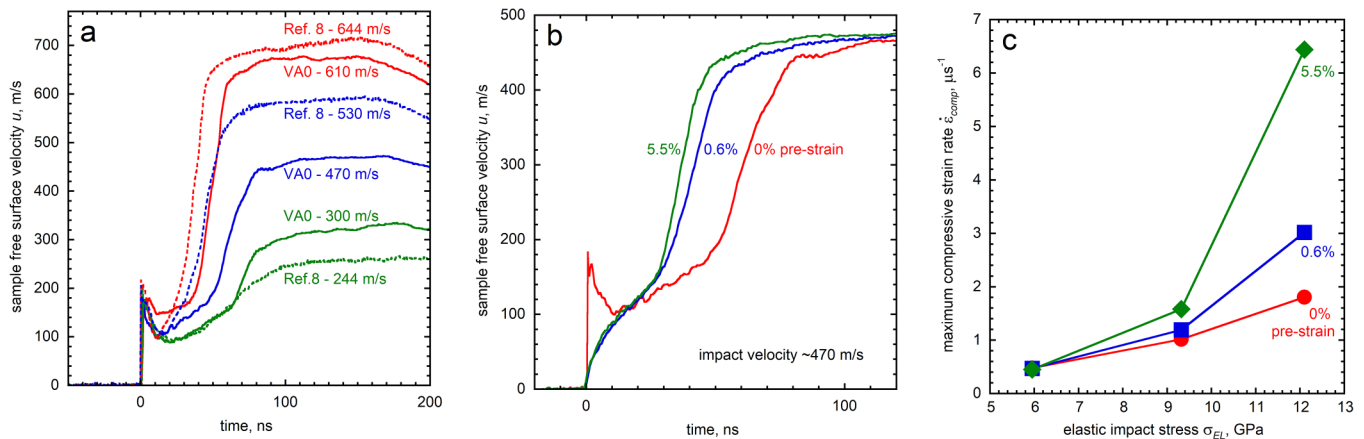
noted that the only change introduced by pre-straining into annealed vanadium is a surplus of the dislocation density, see Subsection III A.

The material's Hugoniot elastic limit,  $\sigma_{HEl}$ , and its dynamic tensile (spall) strength  $\sigma_{sp}$  are routinely estimated based on experimentally obtained velocity histories using Eq. (2a) for the former, and expression<sup>17</sup> (Ref. 33, p. 99)

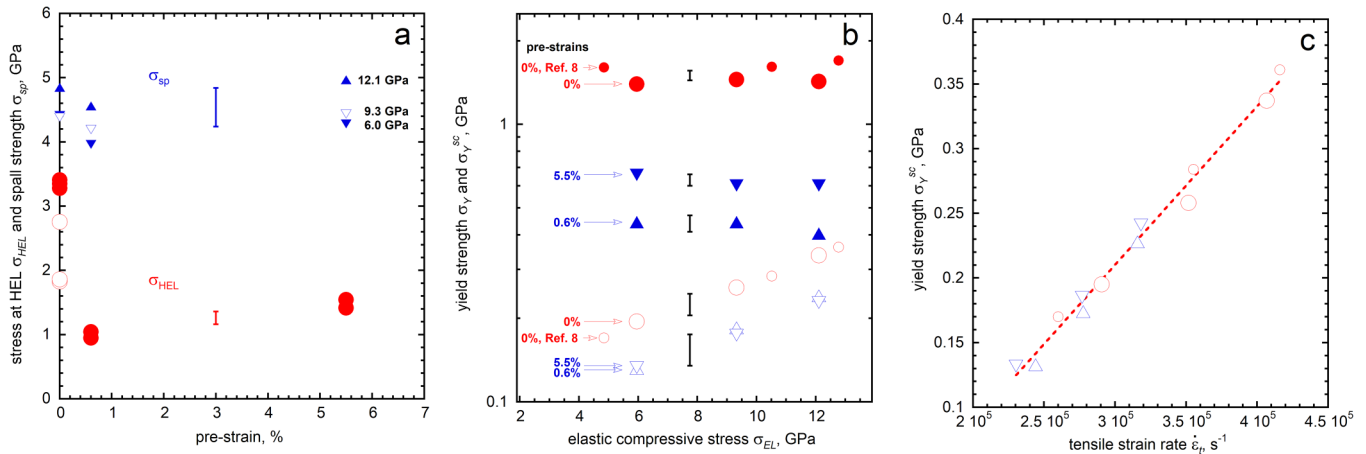
$$\sigma_{sp} = \frac{1}{2} \rho_0 c_b (\Delta u_{pb} + \delta u) \quad (3)$$

for the latter. Here,  $\Delta u_{pb}$  is the velocity pull-back,  $\delta u = \delta h(1/c_b - 1/c_l)|\dot{u}_1|/(\dot{u}_1 + \dot{u}_2)$  is a correction for the distortion of the waveform as a result of the different propagation velocities of the spall signal front and of the rarefaction wave ahead of it in the elasto-plastic media,  $\dot{u}_1$  and  $\dot{u}_2$  are the slopes of the descending and ascending branches of the pull-back signal, respectively, and  $\delta h$  is the thickness of the spall plate determined based on the period  $\delta t_r = 2\delta h/c_l$  of the post-spall reverberation of the part of the momentum in the spall plate, see Fig. 3(a). These,  $\sigma_{HEl}$  and  $\sigma_{sp}$ , material parameters are shown in Fig. 5(a) as a function of

22 December 2025 18:07:02



**FIG. 4.** (a) Front parts of velocity histories obtained with 2 mm thick annealed vanadium samples in the present study (waveforms VA, solid lines) and in Ref. 8 (dashed lines). (b) Front parts of the velocity histories recorded after loading of 2 mm thick differently pre-strained vanadium samples by 0.5 mm thick copper impactors having velocities  $470 \pm 10$  m/s. (c) Maximum compressive strain rate  $\dot{\epsilon}$  as a function of elastic compressive stress  $\sigma_{EL}$ . The size of the symbols corresponds approximately to the uncertainty of the  $\dot{\epsilon}$  determination.



**FIG. 5.** (a) Variation in stress at HEL  $\sigma_{HEL}$  (filled circles) and spall strength  $\sigma_{sp}$  (triangles) of vanadium samples with pre-straining. Different types of triangles correspond to the elastic compressive stress  $\sigma_{EL}$  ranging from 5.9 to 12.1 GPa. The open circles correspond to the  $\sigma_{HEL}$  values determined based on the post-spike velocity minima  $u_{HEL\min}$  [Fig. 3(a)]. (b) Yield stress  $\sigma_Y$  (filled symbols) and  $\sigma_Y^{sc}$  (open symbols) of differently pre-strained vanadium samples as a function of elastic compressive stress  $\sigma_{EL}$ . The small circles correspond to the values of  $\sigma_Y$  and  $\sigma_Y^{sc}$  estimated based on the velocity histories of 2 mm vanadium samples from Ref. 8. The error bars in (a) and (b) correspond to the typical uncertainty of parameter determination. (c) Correlation between the yield strength  $\sigma_Y^{sc}$  of shock-compressed vanadium with tensile strain rate  $\dot{\epsilon}_t$ . Designations are the same as in (b), and the dashed line is the linear fit based on the four types of data.

pre-straining. The strong effect of the latter on  $\sigma_{HEL}$  is evident; a modest, 0.6%, pre-straining it seems changes the mechanism responsible for the onset of plastic deformation. The increase in pre-straining from 0.6% to 5.5% leads to an increase in  $\sigma_{HEL}$  from 1 to 1.5 GPa caused by strain (work) hardening; the 0.6% residual strain was obtained after releasing the sample from 375 MPa compression, while for 5.5% pre-straining, vanadium was compressed up to 625 MPa. The variation in the spall strength  $\sigma_{sp}$  of vanadium with pre-straining is much weaker and does not exceed the uncertainty of  $\sigma_{sp}$  determination.

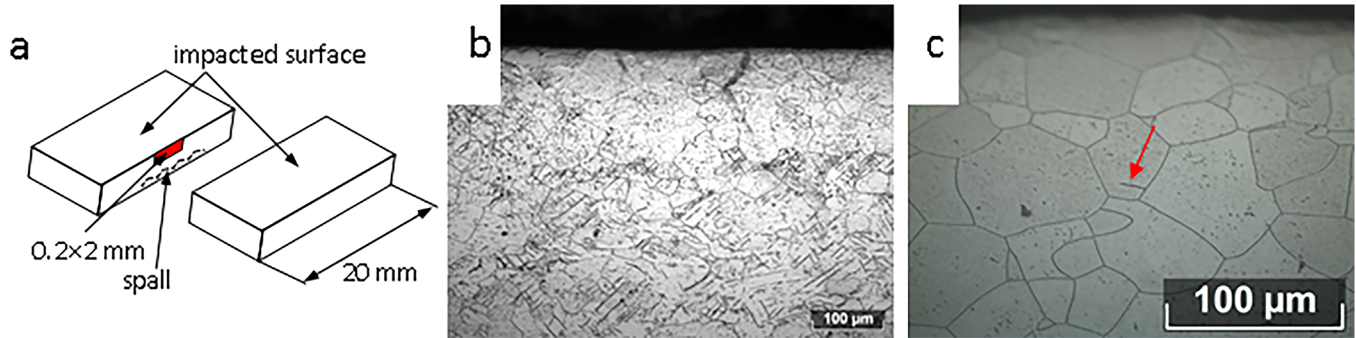
As pointed out in Sec. I, Eq. (1) made it possible to estimate the yield strength  $\sigma_Y^{sc}$  of the shock-compressed material based on the unloading part of its velocity history  $u(t)$ . According to Ref. 19, due to a hysteresis of elasto-plastic compression, the value of particle velocity  $u_{p1}$  is related to free surface velocity  $u_1$  at the beginning of the unloading as  $u_{p1} = (u_1 + \delta_{ep})/2$ , where  $\delta_{ep} = \sigma_{HEL}(1/c_b - 1/c_l)/\rho_0$ , while due to the interaction of the incident and reflected rarefaction waves,  $u_{p2} = u_2/2$ . The dependence  $a(u_p)$  of phase velocity on particle velocity is determined by linear interpolation between points  $[u_{p1}, a_l(u_{p1}) = a_b(u_{p1})c_l/c_0]$  and  $[u_{p2}, a_b(u_{p2}) = c_0 + 2su_{p2}]$ , where  $c_0$  and  $s$  are parameters of linear material's Hugoniot  $U_s = c_0 + su$ . In the present estimations of  $\sigma_Y^{sc}$ , the values of  $c_l = 6090$  m/s,  $\rho_0 = 6110$  kg/m<sup>3</sup> (Table I), and vanadium Hugoniot  $U_s = 5044 + 1.242u$ <sup>34</sup> were used together with the recorded waveforms [an example of the determination of points  $u_1$  and  $u_2$  is shown in Fig. 3(b)]. The values of  $\sigma_Y$  determined based on Eq. (2) and those of  $\sigma_Y^{sc}$  determined using the approach<sup>19</sup> in Eq. (1) are shown in Fig. 5(b), together with  $\sigma_Y$  and  $\sigma_Y^{sc}$  estimated based on the velocity histories of the 2 mm thick samples of the annealed vanadium from Ref. 8. The values  $\sigma_Y$  are almost independent of impact stress and are evidently defined by preexisting dislocation density

$\eta$ , which, in turn, leads to some work hardening. At the same time, both virtually coinciding  $\sigma_Y^{sc}$  values of 0.6% and 5.5% pre-strained samples and those of undeformed (which are greater by some 40%) grow with elastic compressive stress  $\sigma_{EL}$ . The  $\sigma_Y^{sc}$  values correlate with the rate of tensile strain  $\dot{\epsilon}_t$ , which is roughly estimated based on the slopes  $\dot{u}_1$  of the unloading parts of the velocity histories [Fig. 3(a)] as  $\dot{\epsilon}_t \approx \dot{u}_1/2c_b$ . It is noteworthy that all four groups of material follow a single  $\sigma_Y^{sc}(\dot{\epsilon}_t)$  dependence, Fig. 5(c). Since the tensile strain rate during plastic unloading is close to the plastic strain rate, it is likely that both  $\sigma_Y^{sc}$  and  $\dot{\epsilon}_t$  are controlled by the same mechanism.

### C. Effect of the initial sample state on twinning

The samples recovered after impact loading were cut along their centerline, ground, polished, and etched. The count of twins was performed over a  $2 \times 0.2$  mm<sup>2</sup> rectangle adjacent on its long side to the sample impacted surface, as shown in Fig. 6(a). The light microscope image of such a part of the undeformed vanadium sample recovered after 610-m/s impact loading contains a noticeable number of deformation twins, Fig. 6(b). The twins are well developed and usually cross the grain from one side to the other. The count of the twins in this sample gave a twin number of about 1100. The corresponding segments of the cross sections of undeformed samples underwent a weaker, 470 and 300-m/s, loading containing about 750 and 320 twins, respectively. The twin harvesting in the pre-strained samples was much less successful. The twinning was found to be suppressed completely in all 5.5% pre-strained samples.

In the case of 0.6% pre-straining, a single underdeveloped twin was found in the sample recovered after the strongest,



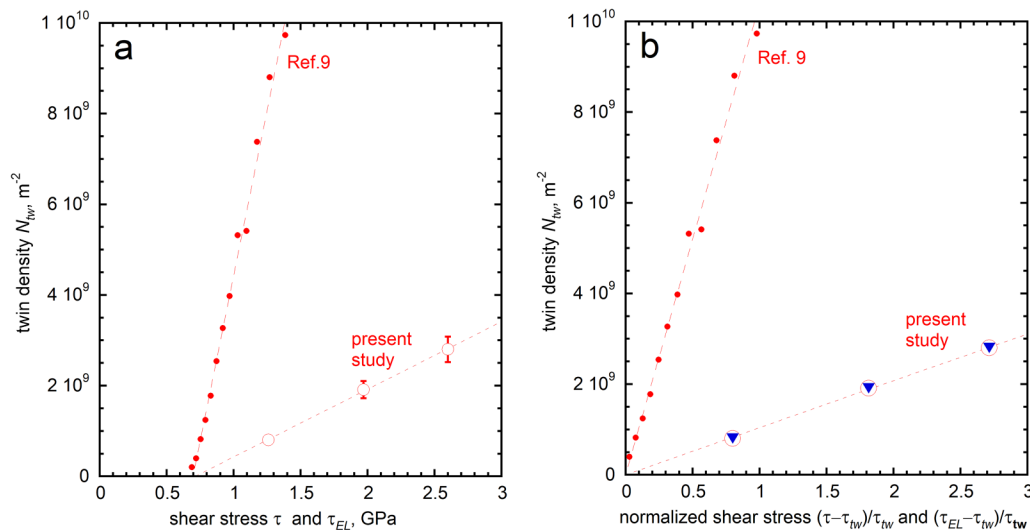
**FIG. 6.** (a) Twin count region:  $2 \times 0.2 \text{ mm}^2$  rectangle around the cross-sectional centerline of a symmetrically cut  $20 \times 20 \text{ mm}^2$  square sample. (b) LM ( $\times 200$ ) image close to the centerline and adjacent to the impacted surface part of an un-pre-strained vanadium sample recovered after loading by a copper impactor with a velocity of about 610 m/s. (c) LM ( $\times 500$ ) image close to the centerline and adjacent to the impacted surface part of a 0.6% pre-strained vanadium sample recovered after the same loading. The arrow shows the only twin found in the sample.

610 m/s, loading, Fig. 6(c). The weaker loaded samples contained no twins.

The results of the present determination of the twin density  $N_{tw}$  (the number of twins per unit area) in the undeformed, VA0-300, VA0-470, and VA0-610, vanadium samples are shown in Fig. 7(a) as a function of shear stress  $\tau_{EL} = \sigma_{EL}G/E'$  corresponding to the elastic compressive stress  $\sigma_{EL}$  at the impactor/sample interface. The same plot includes the dependence  $N_{tw}(\tau)$  obtained in Ref. 9 based on two separate, twin calibration and stress calibration, series of impact experiments with undeformed vanadium samples.

It should be noted that dependencies  $N_{tw}(\tau)$  obtained both in Ref. 9 and in the present study may be somewhat distorted due to detwinning caused by the arrival of the release waves from the perimeter of the shocked sample. Moreover, some additional distortion of  $N_{tw}(\tau)$  dependencies may be related to LM invisible, too small, twins.

There are two features that characterize the two,  $N_{tw}(\tau_{EL})$  and  $N_{tw}(\tau)$ , dependencies. First, the interpolating fit of  $N_{tw}(\tau)$  points crosses the abscissa at  $\tau = 0.69 \text{ GPa}$  and an extrapolation of the  $N_{tw}(\tau_{EL})$  fit crosses the abscissa ( $N_{tw} = 0$ ) almost at the same point,



**FIG. 7.** (a) Presently determined near-surface density  $N_{tw}$  of deformation twins as a function of the shear stress  $\tau_{EL}$  at the impactor/sample interface (large circles) and the values of  $N_{tw}(\tau)$  obtained in Ref. 9 based on separately determined dependencies of  $N_{tw}(h)$  and  $\tau(h)$  on the distance  $h$  from the impactor/sample interface (small circles). The dashed lines correspond to linear fits for  $N_{tw}(\tau_{EL})$  and  $N_{tw}(\tau)$ . (b) The same twin densities  $N_{tw}$  as a function of normalized shear stress  $(\tau - \tau_{tw})/\tau_{tw}$  (filled circles, Ref. 9) and  $(\tau_{EL} - \tau_{tw})/\tau_{tw}$  (open circles, present study). The filled triangles correspond to the shear stress at  $h = 0.1 \text{ mm}$  from the impacted surface estimated based on the shear stress  $\tau(h)$  calibration performed in Ref. 9.

$\tau_{EL} = 0.71$  GPa. It was shown in Ref. 9 that the shear stress at which  $N_{tw}$  became equal to zero corresponded to the material twinning stress, i.e., the threshold stress above which twinning is activated. Consequently, two approaches, that of Ref. 9 and that of the present study, give virtually the same value of the twinning stress in annealed vanadium,  $\tau_{tw} \approx 0.7$  GPa. Second, the slopes of the two dependencies differ by an order of magnitude. This difference, as well as the complete suppression of twinning in the pre-strained samples, is discussed in Sec. IV.

#### IV. DISCUSSION

As was first observed in Refs. 35 and 36, the twins are initiated at the grain boundaries. Both for vanadium studied in Ref. 9 and for that of the present work (see Fig. 6), it is likely to be so. Even the only twin visible in Fig. 6(c) was initiated at the grain boundary. The twin density dependencies  $N_{tw}(\tau)$  determined in Ref. 9 were analyzed based on a recently developed theory,<sup>37,38</sup> which considers twin nucleation as a stress-governed Poisson process of emitting, as partials, the grain boundary dislocations (GBDs) from the grain boundaries. For an average number of twin nuclei  $\langle n(A) \rangle$  generated at the grain boundary area  $A$ , the theory suggests the expression

$$\langle n(A) \rangle = \frac{A}{A_0} \left( \frac{\tau - \tau_{tw}}{\tau_{tw}} \right)^\alpha, \quad (4)$$

where  $(\tau - \tau_{tw})/\tau_{tw}$  is the normalized driving force of twins' nucleation (no twin appear at  $\tau < \tau_{tw}$ ),  $A_0$  is a characteristic grain boundary area at which nucleation of a single twin is expected, and  $\alpha$  is a material parameter. Equation (4) gives for the number  $N_{tw}$  of twin nuclei per unit area,

$$N_{tw} = \frac{\langle n(A) \rangle}{A} = \frac{1}{A_0} \left( \frac{\tau - \tau_{tw}}{\tau_{tw}} \right)^\alpha. \quad (4a)$$

It should be noted that the number of twin nuclei  $N_{tw}$  per unit area in Eq. (4a) and the presently measured numbers of twins  $N_{tw}$  per unit area are identical. As soon as a twin is nucleated, it grows under acting shear stress and the stress required for the growth is usually lower than that required for the nucleation (Ref. 39, p. 60). As was shown in Ref. 9, the twins grow with the velocity that by an order of magnitude is greater than the velocity of dislocations under the same stress.

Accepting for both the vanadium studied in Ref. 9 and for the presently studied vanadium  $\tau_{tw} = 0.7$  GPa, the dependencies  $N_{tw}(\tau)$ , shown in Fig. 7(a), may be replotted using Eq. (4a) with  $\alpha = 1$ , Fig. 7(b). It is noteworthy that the presently performed twin counts for Fig. 7 dealt with the samples' cross-sectional strips bounded by  $h = 0$  (impacted surface) and  $h = 0.2$  mm. Consequently, the reference point for the twin density  $N_{tw}$  should be  $h = 0.1$  mm and the shear stress must be estimated for this point. Based on shear stress calibration  $\tau(h)$  used in Ref. 9, the shear stress  $\tau_{EL}$  determined for  $h = 0$  was corrected for the distance  $h = 0.1$  mm. It is apparent from Fig. 7(b) that after correction, the  $N_{tw}(\tau_{EL})$  dependence stays virtually unchanged.

The slopes of the fitting lines  $N_{tw}(\tau)$  are inversely proportional to the grain boundary area  $A_0$  at which nucleation of a single twin may be expected. From Fig. 7(b), it is evident at a glance that the values of  $A_0$  obtained with two types of vanadium differ by an order of magnitude: the fits give for vanadium studied in Ref. 9,  $A_0 \approx 9 \times 10^{-11} \text{ m}^{-2}$ , while for the presently studied material  $A_0 \approx 9 \times 10^{-10} \text{ m}^{-2}$ . This means that in the case of the presently studied vanadium, the area of the grain surface for which appearance of a single twin nucleus may be expected is by an order of magnitude greater than in the vanadium studied in Ref. 9. It is known that twinning is generally more likely in larger grain-sized materials<sup>38</sup> (and the references therein). At the same time, such suppression of twin productivity of grain boundaries can be attributed to substantial vanadium contamination by both metallic (Si—2000 ppm, Al—2000 ppm) and non-metallic (e.g., C—300 ppm, see Subsection II A) atoms. It seems reasonable that during 1000 °C annealing, metallic atoms (substitutional ones) do not leave the grain bulk, while it is not so with interstitial (C, N, and O) atoms. Carbon, for example, should diffuse from the bulk toward the grain boundaries, creating at the boundaries, VC nano-precipitates.<sup>40</sup> Considering the relatively high diffusion coefficient of carbon in vanadium at 1000 °C,  $D_{C-V} \approx 10^{-7} \text{ cm}^2\text{s}^{-1}$ ,<sup>41</sup> the 1-h annealing is enough to cause leaving the grains' bulk by carbon atoms (as well as by other interstitial atoms such as N and O) and populating the grain boundaries by nano-carbides (nitrides, oxides). Most likely the grain boundaries and part of etch pits in Fig. 2 are decorated by such nano-precipitates. Taking into account the fact that the twinning stress is the same,  $\tau_{tw} \approx 0.7$  GPa, in both materials, it is reasonable to assume that the contaminants do not change the mechanism of twins' nucleation but only occupy the sites which, in cleaner material, may well serve as the sources of the nucleation.

The difference between the densities  $N_{tw}^{cl}$  and  $N_{tw}^{uncl}$  of the twins generated by the same stress in the undeformed vanadium samples of "clean" and "unclean" types, respectively, allows one to conclude that the effect of twinning on its dynamic strength is inessential. Over a wide, 4–12 GPa, range of elastic compressive stresses  $\sigma_{EL}$ , the difference between the values of  $\sigma_Y$  of the two materials does not exceed the uncertainty of  $\sigma_Y$  determination [ $5.5 \pm 0.5$  GPa, see Fig. 5(b)]. From the same Fig. 5(b), it is apparent that the values of  $\sigma_Y^{sc}$  of the undeformed samples also virtually coincide.

The contribution of twins in plastic deformation may be estimated based on the volume fraction of twins  $v_{tw}$  multiplied by modulus  $|s|$  of twinning shear, which is equal to  $\sqrt{2}/2 = 0.707$  in the case of  $\{111\}\{211\}$  twinning. In the present work, the values of  $v_{tw}$  were not determined, but they were measured in Ref. 9, where the highest,  $v_{tw} \sim 0.1$ , value was found for the strongest, 600 m/s, loading. Thus, the twin-related fraction of plastic deformation in Ref. 9 was approximately 0.07. Consequently, in the present study, it cannot be greater than 0.007; i.e., in both types of vanadium, virtually all plastic deformation is realized by the glide. The presence of twins in shock-loaded material, especially in the vicinity of its impacted surface, is likely caused by a relatively high, 1.2–2.5 GPa, shear stress  $\tau_{EL}$  generated at the surface (see Fig. 7) This stress is greater than both the twinning stress,  $\tau_{tw} \approx 0.7$  GPa, and the shear stress  $\tau_Y = \sigma_Y/2$  corresponding to the onset of the dislocation glide under unidimensional quasi-static loading. Based on



$\sigma_Y^{0.1} \approx 0.16$  GPa for the presently studied vanadium (see Subsection II A), the value  $\tau_Y \approx 0.08$  GPa may be accepted. It should be noted that a simultaneous triggering of both twinning and gliding was observed in strongly<sup>42–46</sup> and modestly<sup>9,47</sup> shock-loaded BCC metals and strongly loaded FCC metals<sup>48,49</sup> (and the references therein).

The absence of twins in the pre-strained samples and their modest presence in the undeformed vanadium samples recovered after impact loading allows us to assume that the differences between the compressive parts of the waveforms shown in Fig. 4(b) originate only from the difference in the samples' initial structure—the difference in dislocation density  $\eta$ . This assumption is justified by Fig. 8, where velocity histories recorded after 300 m/s loading of the undeformed and pre-strained vanadium samples are compared with the results of the unidimensional finite difference (1- $\mu$  Lagrangian grid) numerical simulation of these tests using a dislocation-based constitutive model for vanadium (Ref. 50, p. 54). The density of the mobile dislocations in this model is related to the plastic strain  $\gamma$  as  $\eta = \eta_0 + a\gamma$ , where  $\eta_0$  is the initial density of the mobile dislocations and  $a$  is a material constant,<sup>51</sup> while the plastic strain rate is determined by using the Orowan equation  $\dot{\gamma} = \eta b \bar{v}$ , where  $b = 0.262$  nm is the Burgers vector of the vanadium lattice and  $\bar{v}$  is the average dislocation velocity. For the latter, the expression  $\bar{v} = c \exp[-D/(\sigma_{eq} - \sigma_{min})]$  is suggested in Ref. 51. With an increase in equivalent stress  $\sigma_{eq}$  (in the present case,

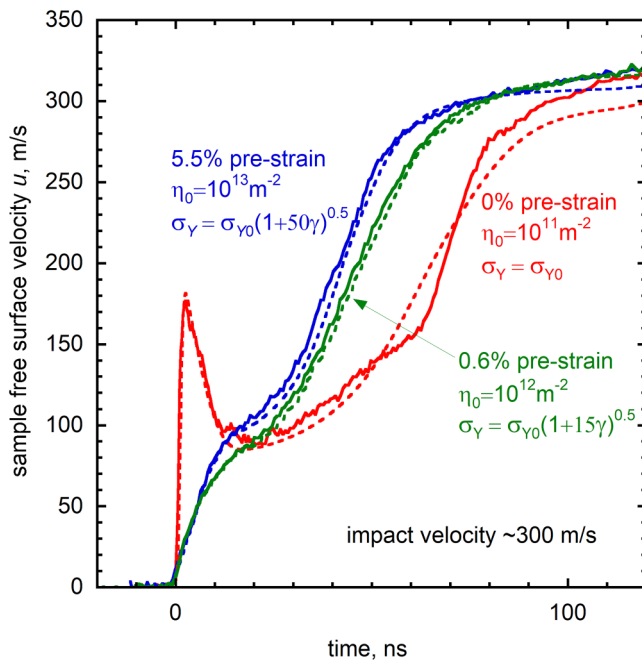
$\sigma_{min} = 0$ ), the velocity of the dislocations approaches the shear speed of sound  $c = c_s = 2830$  m/s in vanadium. The remaining constants,  $D = 1.3 \times 10^9$  Pa and  $a = 8.5 \times 10^{14}$ , were chosen to fit reasonably well the velocity history of the un-pre-strained sample. The work hardening is described as  $\sigma_Y = \sigma_{Y0}(1 + \beta\gamma)^n$ ,<sup>52</sup> where  $\sigma_{Y0}$  is taken to be equal to  $\sigma_Y^{0.1} = 0.16$  GPa (see Subsection II A). All, except two, vanadium constitutive parameters in the two simulations were held unchanged, while the initial densities  $\eta_0$  of the mobile dislocations and work hardening parameter  $\beta$  were taken to be equal to  $\eta_0 = 10^{11} \text{ m}^{-2}$ ,  $\eta_0 = 10^{12} \text{ m}^{-2}$ , and  $\eta_0 = 10^{13} \text{ m}^{-2}$  and  $\beta = 0$ ,  $\beta = 15$ , and  $\beta = 50$  for the undeformed, 0.6%, and 5.5% pre-strained material, respectively. A reasonable outcome of the simulations is that the vast majority of the dislocations created by pre-straining are mobile.

Consequently, the initial, created by pre-straining, densities of the mobile dislocations  $\eta_0$ , together with modest work hardening, are responsible for establishing the front part of the velocity histories. Due to a relatively low  $\eta_0$ , the decrease in  $\sigma_{HEL}$  (and also in  $\sigma_Y$ ) with respect to its initial value  $\sigma_{EL}$  and the steepening of the front of the plastic wave after a 2 mm traverse over pristine material are much less pronounced in the undeformed samples [Fig. 5(b)]. The lack of  $\eta_0$  remains noticeable even in the shock-compressed state; the  $\sigma_Y^{sc}$  of the pre-strained samples is 1.5 times lower than that of undeformed samples.

## V. CONCLUSIONS

The effect of modest, 0.6% and 5.5%, pre-straining on the impact response of 2 mm thick samples of annealed polycrystalline vanadium of commercial purity was studied in a series of VISAR-instrumented planar impact tests in which the samples were recovered after loading by 0.5 mm thick copper impactors having velocities of about 300, 470, and 610 m/s. Based on recorded velocity histories, the strength of the samples in compression,  $\sigma_Y$ , and in the shock-compressed state,  $\sigma_Y^{sc}$ , was determined. The  $2 \times 0.2 \text{ mm}^2$  strips of the cross sections of the recovered samples adjacent to their impacted surface were analyzed for the presence of shock-induced twins, and the number of twins  $N_{tw}$  per unit area was determined. The outcomes of such combined shock wave experiments are as follows:

1. Shock-induced twins are found only in shock-loaded undeformed samples. Both 0.6% and 5.5% pre-straining suppress twinning completely.
2. The dependence of  $N_{tw}$  on shear stress  $\tau_{EL}$  corresponding to elastic compressive stress  $\sigma_{EL}$  is linear. This result is in line with the probabilistic theory of stress-induced twin nucleation from grain boundary dislocations.<sup>37,38</sup> A linear extrapolation of  $N_{tw}(\tau_{EL})$  to  $N_{tw} = 0$  gives the twinning stress (the stress below which no twinning is possible)  $\tau_{tw} \approx 0.7$  GPa, which is very close to  $\tau_{tw} = 0.69 \pm 0.03$  GPa found in Ref. 9.
3. It is found that the presence of impurity atoms, rather non-metallic, such as C, N, and O, may decrease substantially the number (and density) of shock-induced twins. The foreign atoms do not change the mechanism of twinning nucleation but decrease the number of nucleation sites at the grain boundaries by occupying a part of them with carbide/nitride/oxide nano-precipitates.



**FIG. 8.** Experimentally recorded velocity histories of undeformed and 0.6% and 5.5% pre-strained vanadium samples impact loaded with 300 m/s velocity (solid lines) and the corresponding numerically simulated waveforms (dashed lines) obtained with different values of initial dislocation density  $\eta_0$  and different strengthening coefficients  $\beta$ .

4. It is shown that despite the presence of twins, their contribution in the plastic deformation of shocked undeformed vanadium samples is very modest. In the pre-strained samples, the plastic deformation occurs by glide only.
5. The main parameter governing the strength  $\sigma_Y$  of pristine (in the shock sense) material is the initial dislocation density  $\eta_0$ . This statement is supported strongly by numerical simulations using a dislocation-based description of vanadium plasticity. It is noteworthy that the initial  $\eta_0$  also affects the vanadium's strength  $\sigma_Y^c$  in the shock-compressed state, i.e., the shock-loaded material "remembers" its pre-shock state.

The twinning stress presently found from an extrapolation of  $N_{tw}(\tau_{EL})$  dependence to  $N_{tw} = 0$  virtually coincides with the twinning stress determined in Ref. 9 based on two types of impact tests: the stress calibration tests providing dependence  $\tau(h)$  of the shear stress on the distance  $h$  from the impacted surface and twin calibration tests giving  $N_{tw}(h)$  dependence. Taking into account that the presently used approach requires much less experimental effort, the currently suggested technique may serve as a useful tool for the determination of  $\tau_{tw}$  in different materials. Applying the described method of establishing  $N_{tw}(\tau_{EL})$  and  $\tau_{tw}$  to the study of twinning in solids at different (either pure or gradually contaminated) initial states will evidently deepen our understanding of the processes of twin nucleation.

## ACKNOWLEDGMENTS

Shock wave experiments were performed using the equipment of the Moscow Regional Explosion Center of Russian Academy of Sciences. Financial support from the Russian Foundation Basic Research (Grant No. 19-02-00416A), from the Ministry of Science and Higher Education of the Russian Federation (State Assignment Nos. AAA-A19-119071190040-5 and 075-00460-21-00) (S. V. Razorenov, A. S. Savinykh, and G. V. Garkushin), and from the Israel Science Foundation (Grant No. 441/18) (E. B. Zaretsky and N. Frage) are gratefully acknowledged.

## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## REFERENCES

- <sup>1</sup>T. G. Lindley and R. E. Smallman, "The plastic deformation of polycrystalline vanadium at low temperatures," *Acta Metall.* **11**(5), 361–371 (1963).
- <sup>2</sup>J. Bressers, M. Heerschap, and P. De Meester, "Deformation properties of vanadium single crystal," *J. Less Common Met.* **22**(3), 321–326 (1970).
- <sup>3</sup>G. Li-Jiang, L. Yue-Lin, W. Wei-Tian, Z. Ying, L. Guang-Hong, and Y. Jun-En, "Effects of chromium on structure and mechanical properties of vanadium: A first-principles study," *Chin. Phys. B* **22**, 106109 (2013).
- <sup>4</sup>L. H. Yang, M. Tang, and J. A. Moriarty, "Chapter 92: Dislocations and plasticity in bcc transition metals at high pressure," *Dislocation. Solids* **16**, 1–46 (2010).
- <sup>5</sup>J. W. Edington, T. C. Lindley, and R. E. Smallman, "Strain-ageing of vanadium," *Acta Metall.* **12**(9), 1025–1031 (1964).
- <sup>6</sup>A. Dilawar, M. Nasreen, and M. Z. Butt, "Investigation of active slip-systems in some body-centered cubic metals," *J. Mater. Sci.* **46**(11), 3812–3821 (2011).
- <sup>7</sup>Y. Lu, X. Shu, and X. Liao, "Size effect for achieving high mechanical performance body-centered cubic metals and alloys," *Sci. China Mater.* **61**(12), 1495–1516 (2018).
- <sup>8</sup>E. B. Zaretsky and G. I. Kanel, "Tantalum and vanadium response to shock-wave loading at normal and elevated temperatures: Non-monotonous decay of the elastic wave in vanadium," *J. Appl. Phys.* **115**, 243502 (2014).
- <sup>9</sup>A. Sher, L. Meshi, S. Kalabukhov, N. Frage, and E. B. Zaretsky, "Shock wave determination of temperature dependence of twinning stress in vanadium and tantalum," *Mater. Sci. Eng. A* **833**, 142537 (2022).
- <sup>10</sup>B. A. Loomis, "Brittle-ductile transition in vanadium," Retrospective theses and dissertations, Iowa State University, 1958, see <https://lib.dr.iastate.edu/rtd/2258>.
- <sup>11</sup>J. W. Edington and R. E. Smallman, "On mechanical twinning in single crystals of vanadium," *Acta Metallurgica* **13**(7), 765–770 (1965).
- <sup>12</sup>S. Nemat-Nasser and W. Guo, "High strain-rate response of commercially pure vanadium," *Mech. Mater.* **32**(4), 243–260 (2000).
- <sup>13</sup>R. Gröger, Z. Chlup, and T. Kuběnová, "Deformation twinning in vanadium single crystals tested in compression at 77 K," *Mater. Sci. Eng. A* **737**(8), 413–421 (2018).
- <sup>14</sup>M. A. Meyers, O. Vöhringer, and V. A. Lubarda, "The onset of twinning in metals: A constitutive description," *Acta Mater.* **49**, 4025–4039 (2001).
- <sup>15</sup>S. Kibey, J. B. Liu, D. D. Johnson, and H. Sehitoglu, "Predicting twinning stress in fcc metals: Linking twin-energy pathways to twin nucleation," *Acta Mater.* **55**, 6843–6851 (2007).
- <sup>16</sup>J. Wang and H. Sehitoglu, "Twinning stress in shape memory alloys: Theory and experiments," *Acta Mater.* **61**, 6790–6801 (2013).
- <sup>17</sup>T. Antoun, L. Seaman, D. Curran, G. I. Kanel, S. V. Razorenov, and A. V. Utkin, *Spall Fracture* (Springer, New-York, 2002).
- <sup>18</sup>J. L. Brown, C. S. Alexander, J. R. Asay, T. J. Vogler, and J. L. Ding, "Extracting strength from high pressure ramp-release experiments," *J. Appl. Phys.* **114**, 223518 (2013).
- <sup>19</sup>G. I. Kanel and A. S. Savinykh, "Evaluation of the yield stress of solids upon unloading from a shock-compressed state," *Dokl. Phys.* **65**, 12–14 (2020).
- <sup>20</sup>A. N. Dremin and G. I. Kanel, "Compression and rarefaction waves in shock-compressed metals," *J. Appl. Mech. Techn. Phys.* **17**(2), 263–267 (1977).
- <sup>21</sup>G. I. Kanel, S. V. Razorenov, and V. E. Fortov, *Shock-Wave Phenomena and the Properties of Condensed Matter* (Springer, 2004).
- <sup>22</sup>G. I. Kanel, A. S. Savinykh, G. V. Garkushin, and S. V. Razorenov, "Effects of temperature on the flow stress of aluminum in shock waves and rarefaction waves," *J. Appl. Phys.* **127**(3), 035901 (2020).
- <sup>23</sup>G. I. Kanel, A. S. Savinykh, G. V. Garkushin, and S. V. Razorenov, "Effects of temperature and strain on the resistance to high-rate deformation of copper in shock waves," *J. Appl. Phys.* **128**, 115901 (2020).
- <sup>24</sup>G. I. Kanel, A. S. Savinykh, G. V. Garkushin *et al.*, "High-rate deformation of titanium in shock waves at normal and elevated temperatures," *J. Exp. Theor. Phys.* **132**, 438–445 (2021).
- <sup>25</sup>L. M. Barker and R. E. Hollenbach, "Laser interferometer for measuring high velocities of any reflecting surface," *J. Appl. Phys.* **43**, 4669 (1972).
- <sup>26</sup>J. Hirth and J. Lothe, *Theory of Dislocations*, 2nd ed. (Krieger Publishing Company, 1992).
- <sup>27</sup>P. S. DeCarli and M. A. Meyers, "Design of uniaxial strain shock recovery experiments," in *Shock Waves and High-Strain-Rate Phenomena in Metals*, editors by M. A. Meyers and L. E. Murr (Plenum Press, New York London, 1981), Chap. 22, pp. 341–373.
- <sup>28</sup>N. K. Bourne and G. T. Gray, "Computational design of recovery experiments for ductile metals," *Proc. R. Soc. London, A* **461**, 3297–3312 (2005).
- <sup>29</sup>D. L. Higgins, B. Pang, J. C. F. Millett, G. Whiteman, I. P. Jones, and Y.-L. Chiu, "Contrasting the microstructural and mechanical response to shock loading of cold-rolled copper with annealed copper," *Metall. Mater. Trans. A* **46**, 4518–4521 (2015).